

Task 1: Simple Conditional Statements

Problem:

Write a program that takes an integer as input and prints whether the number is positive, negative, or zero.

Example:

Input: 5

Output: Positive number

Task 2: Even or Odd

Problem:

Write a program that takes an integer as input and checks whether the number is even or odd. It should print “Even” or “Odd” based on the input.

Example:

Input: 7

Output: Odd

Task 3: Age Eligibility

Problem:

Write a program that asks for the user’s age and checks if they are eligible to vote. The voting age is 18 or older.

Example:

Input: 19

Output: Eligible to vote

Task 4: Largest of Three Numbers

Problem:

Write a program that takes three numbers as input and prints the largest one. Use conditional statements.

Example:

Input: 3, 9, 5

Output: 9

Task 5: Comparing Two Numbers

Problem:

Write a program that takes two numbers as input and compares them using comparison operators. It should print:

- "Equal" if the numbers are the same
- "Greater" if the first number is greater
- "Smaller" if the first number is smaller

Example:

Input: 7 and 10

Output: Smaller

Task 6: Multiple Conditions (Logical AND/OR)

Problem:

Write a program that takes two inputs representing the current day and the temperature.

- If the day is "Saturday" or "Sunday" and the temperature is above 30°C, it should print "It's a great day for the beach."
- Otherwise, it should print "Not a beach day today."

Example:

Input: Day = Saturday, Temperature = 32

Output: It's a great day for the beach

Task 7: Grade Classification (Using Nested Conditions)

Problem:

Write a program that takes a student's score (between 0 and 100) and classifies it into one of the following grades:

- A (90-100)
- B (80-89)
- C (70-79)
- D (60-69)
- F (Below 60)

Example:

Input: 85

Output: Grade B

Task 8: Leap Year Checker**Problem:**

Write a program that checks whether a given year is a leap year. A leap year:

- Is divisible by 4, but not by 100, unless it is also divisible by 400.

Example:

Input: 2024

Output: Leap Year

Task 9: Discount Eligibility**Problem:**

Write a program that takes the total amount of a purchase and checks if the customer is eligible for a discount:

- If the purchase is greater than \$100, apply a 10% discount.
- Otherwise, print "No discount."

Example:

Input: 120

Output: You get a 10% discount. Final price is \$108.

Task 10: Vowel or Consonant**Problem:**

Write a program that asks the user for a letter and checks if it's a vowel (a, e, i, o, u) or a consonant. It should output the appropriate message.

Example:

Input: e

Output: Vowel

Task 11: BMI Calculator

Problem:

Write a program that calculates the Body Mass Index (BMI) and classifies it:

- Underweight: $\text{BMI} < 18.5$
- Normal weight: $18.5 \leq \text{BMI} < 24.9$
- Overweight: $\text{BMI} \geq 25$

Use the formula:

$$\text{BMI} = \text{weight (kg)} / (\text{height (m)} * \text{height (m)})$$

Example:

Input: Weight = 70, Height = 1.75

Output: Normal weight

Task 12: Number Divisibility

Problem:

Write a program that asks for a number and checks if it is divisible by both 3 and 5.

Example:

Input: 15

Output: Divisible by both 3 and 5

Task 13: User Login System

Problem:

Write a program that simulates a simple user login system. The program should:

- Ask for a username and password.
 - If the username is "user123" and the password is "pass456", print "Login successful".
 - Otherwise, print "Invalid credentials".
-

Task 14: Traffic Light System

Problem:

Write a program that asks for the traffic light color and prints the corresponding message:

- Red → Stop

- Yellow → Get Ready
- Green → Go

Example:

Input: `Green`

Output: `Go`

Task 15: Check Divisibility and Parity**Problem:**

Write a program that asks for a number and checks the following:

- If the number is divisible by 2 (even) or not (odd).
- If the number is divisible by 3.
- It should print the corresponding messages for each condition.

Example:

Input: `6`

Output: `Even number, Divisible by 3`